Understanding Input Selectivity in Mamba: Impact on Approximation Power, Memorization, and Associative Recall Capacity

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Motivation & Our Contribution

Mamba[1], a State Space Model (SSM) rivaling Transformer
- 🌟 Selective S6 layer: Input-dependent SSM parameters
- 🆕 Mixer Architecture: +Convolution, +Gating
- ✅ Performance: memorization, associative recall

Why **Mamba** works so well?

- Due to input selectivity in **the S6 layer**?

☑ S6 can approximate *projections* to *Haar wavelet*
☑ S6 can dynamically memorize

- Credit to other components in the **architecture**?

☑ Convolution and gating components allow for more efficient solutions to MQAR

- **Why** do we care?

☑ Identify the solution mechanism explicitly
☑ Propose improvements

Definitions

SSM: $\dot{\boldsymbol{h}}(t) = \boldsymbol{\Lambda}(t)\,\boldsymbol{h}(t) + \boldsymbol{B}(t)\,x(t), \quad \boldsymbol{y}(t) = \boldsymbol{C}(t)\,\boldsymbol{h}(t)$

Solution: $\boldsymbol{h}(t) = \int_0^t e^{\int_s^t \boldsymbol{\Lambda}(r)\,dr} \boldsymbol{B}(s)\,x(s)\,ds \equiv \langle g, x\rangle$ *(How powerful is the basis g)*

Solution (Discrete): $\boldsymbol{h}(t) = \sum_1^t \left(\prod_{r=s+1}^t \overline{\boldsymbol{\Lambda}}_r\right) \overline{\boldsymbol{B}}_s\, x_s$ *(How to curb memory decay)*

SSM Params: $\overline{\boldsymbol{\Lambda}}_t = e^{\boldsymbol{\Lambda}(t)\Delta(t)}, \quad \overline{\boldsymbol{B}}_t \approx B(t)\Delta(t), \quad \overline{\boldsymbol{C}}_t \approx C(t)$

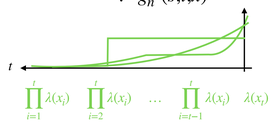
- Mamba: $\boldsymbol{\Lambda}(t) = \boldsymbol{\Lambda}$, $\Delta(t) = \mathrm{Softplus}(\mathrm{Linear}(x_t))$, $B(t) = \mathrm{Linear}(x_t)$, $C(t) = \mathrm{Linear}(x_t)$ *(Input-dependent)*
- S4D: $\boldsymbol{\Lambda}(t) = \boldsymbol{\Lambda}, \Delta(t) = 1, B(t) = B, C(t) = C$

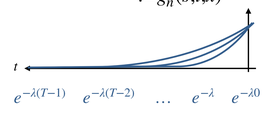
Mamba	Mamba-2
$\boldsymbol{\Lambda} \in \mathbb{R}^{d\times N}, \boldsymbol{x}_t \in \mathbb{R}^d, \boldsymbol{h}_t \in \mathbb{R}^{d\times N}$	$\boldsymbol{\Lambda} = \lambda \in \mathbb{R}, \boldsymbol{x}_t \in \mathbb{R}^d, \boldsymbol{h}_t \in \mathbb{R}^{d\times N}$
$\hat{\boldsymbol{x}}_t = \sigma(\mathrm{conv}(\boldsymbol{x})_{[t]}) \in \mathbb{R}$	$\hat{\boldsymbol{x}}_t = \sigma(\mathrm{conv}_u(\mathrm{Linear}(\boldsymbol{x}))_{[t]}) \in \mathbb{R}^d$
$\Delta_t = \mathrm{SoftPlus}(\mathrm{Linear}(\hat{\boldsymbol{x}}_t)) \in \mathbb{R}^d$ $(\Delta_t^{\mathrm{S4D}} = 1)$	$\Delta_t = \mathrm{SoftPlus}(\mathrm{Linear}(\boldsymbol{x}_t)) \in \mathbb{R}$
$\boldsymbol{B}_t = \mathrm{Linear}(\hat{\boldsymbol{x}}_t) \in \mathbb{R}^N$ $(B_t^{\mathrm{S4D}} = B)$	$\boldsymbol{B}_t = \sigma(\mathrm{conv}_B(\mathrm{Linear}(\boldsymbol{x}))_{[t]}) \in \mathbb{R}^N$
$\boldsymbol{C}_t = \mathrm{Linear}(\hat{\boldsymbol{x}}_t) \in \mathbb{R}^N$ $(C_t^{\mathrm{S4D}} = C)$	$\boldsymbol{C}_t = \sigma(\mathrm{conv}_C(\mathrm{Linear}(\boldsymbol{x}))_{[t]}) \in \mathbb{R}^N$
Convolution	
$\boldsymbol{h}_t = e^{\boldsymbol{\Lambda}\odot(\Delta_t \otimes \mathbf{1}_N)} \odot \boldsymbol{h}_{t-1} + (\Delta_t \odot \hat{\boldsymbol{x}}_t) \otimes \boldsymbol{B}_t$	$\boldsymbol{h}_t = e^{\lambda\Delta_t}\boldsymbol{h}_{t-1} + (\Delta_t \hat{\boldsymbol{x}}_t) \otimes \boldsymbol{B}_t$
SSM	
$\boldsymbol{y}_t = \boldsymbol{h}_t \boldsymbol{C}_t \in \mathbb{R}^d, \quad \tilde{\boldsymbol{y}}_t = g(\boldsymbol{x}_t) \odot \boldsymbol{y}_t := \sigma(\mathrm{Linear}(\boldsymbol{x}_t)) \odot \boldsymbol{y}_t \in \mathbb{R}^d$	Gate

Mamba S6 Layer: Approximation on Wavelets

☑ S6 can approximate *projections* to *Haar wavelet;*
☑ S6 has better approximation error rate than S4D on discontinuous functions.

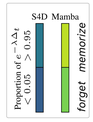
$$h^M(t) = \int_0^t \underbrace{e^{-\lambda_n \int_s^t \Delta(x_r)dr} B_n}_{:=g_n^M(s;t,x)}\, x(s)\,ds; \quad h^S(t) = \int_0^t \underbrace{e^{-\lambda_n(t-s)} B_n}_{:=g_n^S(s;t,x)}\, x(s)\,ds\,.$$

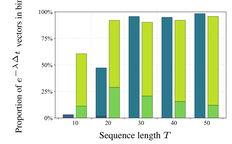




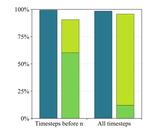
Task Keep-Fifth:
memorize the 5th token in the input sequence

	Mamba+PE	Mamba	S4D	S4D+PE	Transformer
Accuracy↑	**1.00**	0.08	0.09	0.08	**1.0**
Parameters	9.2k	9.3k	8.8k	8.7k	6.4k





$e^{-\lambda\Delta_t} \to 1$ *as length T increases*



S6 can selectively forget and memorize

Mamba S6 Layer: Sensitivity Analysis

$$\frac{\partial \boldsymbol{h}_t}{\partial x_j} = \frac{\partial}{\partial x_j}\left(\sum_{s=1}^{t}\left(\prod_{r=s+1}^{t}\overline{\boldsymbol{\Lambda}}_r\right)\overline{\boldsymbol{B}}_s x_s\right) = \left(\prod_{r=j+1}^{t}\overline{\boldsymbol{\Lambda}}_r\right)\left(\frac{\partial}{\partial x_j}(\overline{\boldsymbol{B}}_j x_j) + \frac{\partial \overline{\boldsymbol{\Lambda}}_j}{\partial x_j}\sum_{s=1}^{j-1}\left(\prod_{r=s+1}^{j-1}\overline{\boldsymbol{\Lambda}}_r\right)\overline{\boldsymbol{B}}_s x_s\right).$$

☑ S6's sensitivity $\partial \boldsymbol{h}_t^M / \partial x_j$ decays *exponentially* unless it "freezes time" via $\lim_{t\to\infty} \lambda_n \sum_{r=1}^{t} \Delta(x_r) \le c$.
☑ Long-range memorization condition: $e^{-\lambda\Delta_t} \to 1$

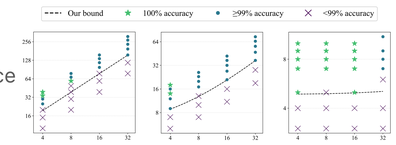
Full Mamba Architecture: 1-layer Mamba Solutions to Associative Recall Tasks

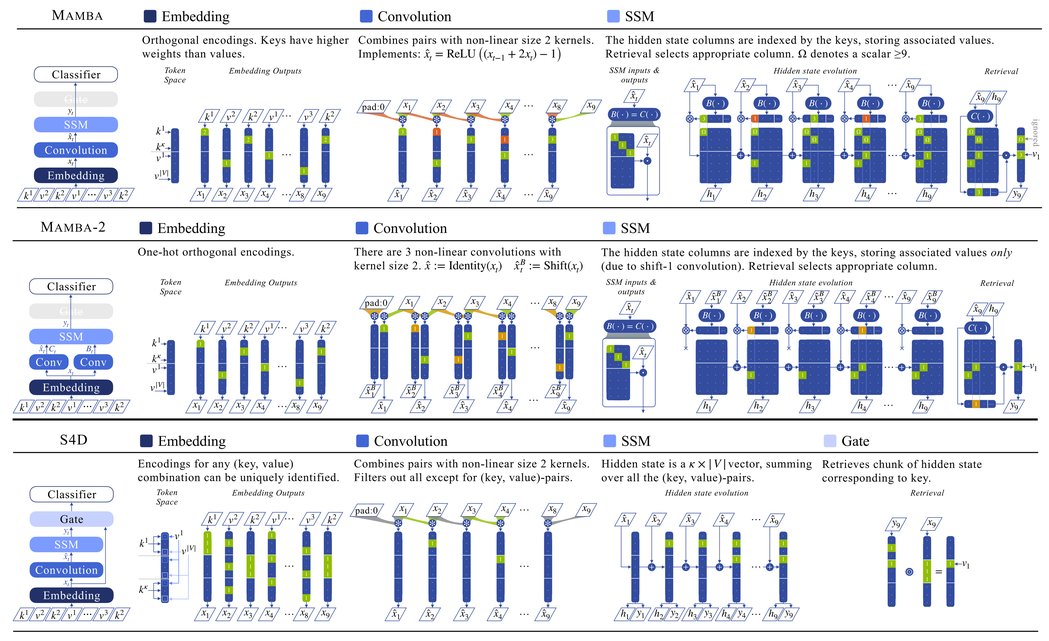
Task Multiple-Query Associative Recall (MQAR)[2]
κ keys, $|V|$ values

☑ Mamba-2 offers *leaner* solution via 3 *convolutions*
☑ *Gated-convolution with S4D* alone can solve MQAR

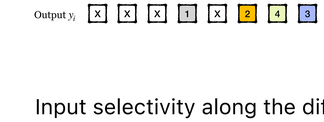
Mixer	Embedding Size	State Size
Mamba	$O(\kappa + \log\lvert V\rvert)$	κ
Mamba-2	$O(\log\kappa + \log\lvert V\rvert)$	$\log\kappa$
S4D	$O(\kappa \log\lvert V\rvert)$	1

Theory ◀ ▶ Practice





Task Induction Heads (IndH)[3]



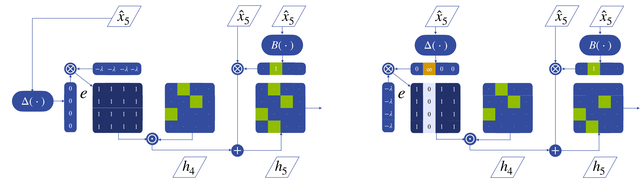
☑ Improve Mamba by incorporating *selectivity differently* in the state matrix, combining insights from sensitivity analysis (top) and MQAR task solution (left)

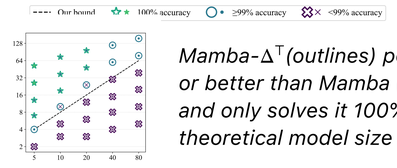
Input selectivity along the different axes of the state matrix $\overline{\boldsymbol{\Lambda}}$

$$\boldsymbol{h}_t = \overline{\boldsymbol{\Lambda}}\,\boldsymbol{h}_{t-1} + \hat{\boldsymbol{x}}_t \otimes \boldsymbol{B}_t$$

Mamba: $\overline{\boldsymbol{\Lambda}} = e^{\boldsymbol{\Lambda}\odot(\Delta(\hat{\boldsymbol{x}}_t)\otimes \mathbf{1}_N)}$ (embedding dimension)

Mamba-Δ^{T}: $\overline{\boldsymbol{\Lambda}} = e^{\boldsymbol{\Lambda}\odot(\mathbf{1}_d \otimes \Delta(\hat{\boldsymbol{x}}_t))}$ (state dimension)





Mamba-Δ^{T}(outlines) performs equally or better than Mamba (filled) on IndH, and only solves it 100% above the theoretical model size (black line)

References

[1] Gu and Dao, Mamba: Linear-Time Sequence Modeling with Selective State Space
[2] Arora, S. et al, Zoology: Measuring and Improving Recall in Efficient Language Models
[3] Sanford, C. et al, One-Layer Transformers Fail to Solve the Induction Heads Task

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